Assignment-1

Verilog Lab Questions

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1 Question 1

Combining Two 4-bit Numbers

```
module combine_bits(
    input [3:0] a,
    input [3:0] b,
    output [7:0] y
    );
    assign y = {a,b};//joins two inputs to one 8-bit output
endmodule
```

Figure 1: question-1 code

```
module tb_combine_bits;

reg  [3:0] a, b; //inputs
wire [7:0] y; //output

combine_bits uut ( .a(a) , .b(b) , .y(y));

initial begin
    $display(" a    b    y");
    $monitor("%b  %b  %b", a, b, y);

   //testing for different cases
    a = 4'b0001; b = 4'b0010; #10;
    a = 4'b1010; b = 4'b0101; #10;
    a = 4'b1111; b = 4'b0000; #10;

    $finish;
end
endmodule
```

Figure 2: testbench-code

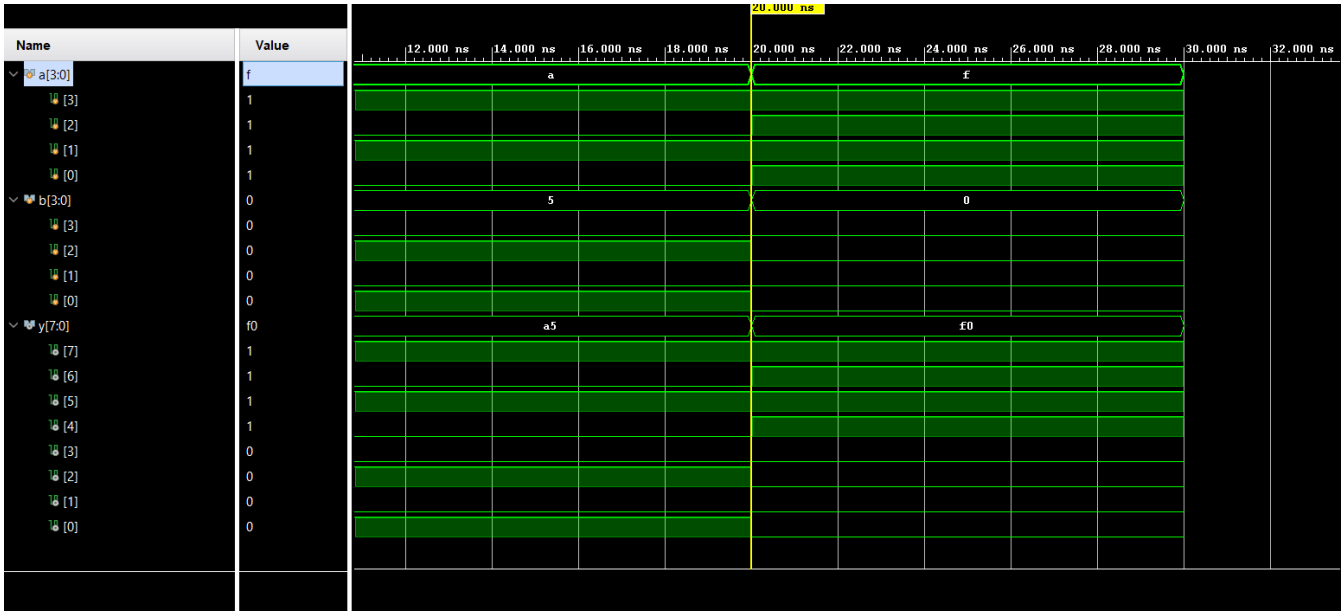

Figure 3: Waveform-1

2 Question 2

Extract Lower Nibble

```
module nibble_extract(
    input [7:0] in,
    output [3:0] out
    );

    assign out = in[3:0];//selects the bits 3 to 0 from the input
endmodule
```

Figure 4: question-2 code

```
module tb_extract;

reg [7:0] in; // input
wire [3:0] out; //output

nibble_extract uut(.in(in) , .out(out) );

initial begin

  $display(" in      out ");
  $monitor("%b    %b",in , out);

  //different test inputs with 5ns delay
  in = 8'b11110001; #5;
  in = 8'b10101011; #5;
  in = 8'b10001000; #5;
  in = 8'b01001101; #5;

 $finish;
 end
 endmodule
```

Figure 5: testbench-code

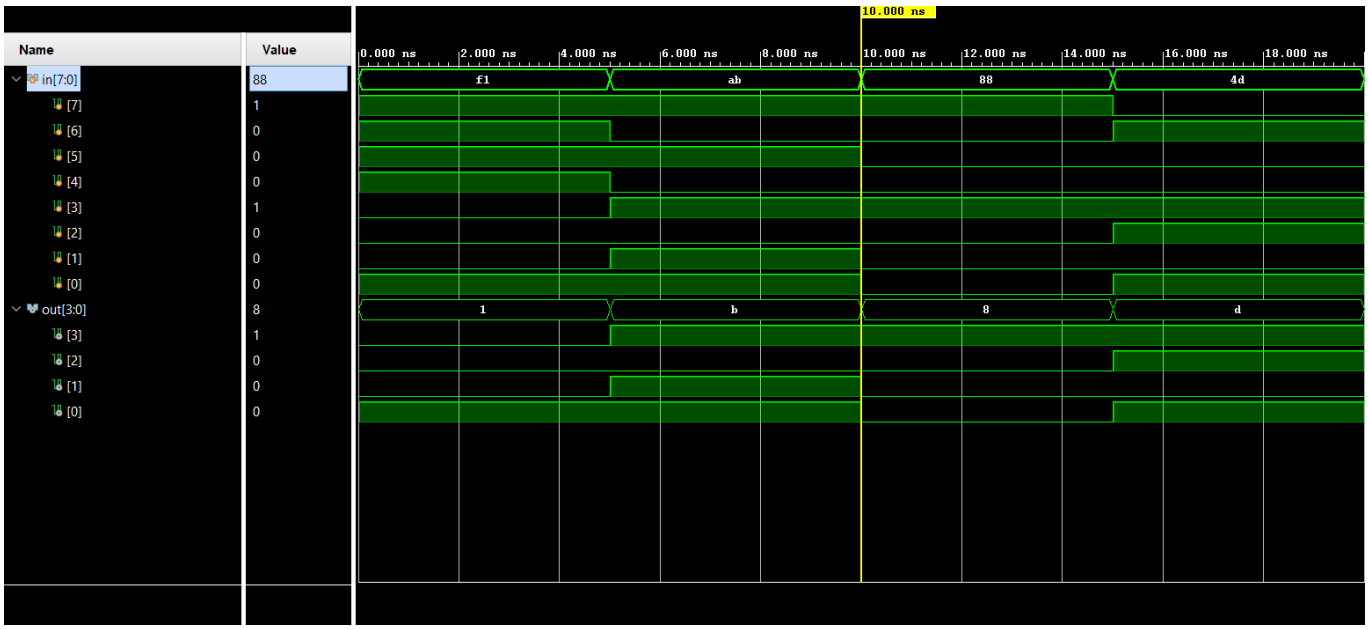

Figure 6: Waveform -2

3 Question 3

Bitwise AND or OR Selector

```
module and_or_selector(
    input [3:0] a,
    input [3:0] b,
    input sel,
    output [3:0] y
    );

    assign y = (sel) ? (a | b) : (a & b);//if sel=0 , y = a&b and if sel=1 , y = a|b
endmodule
```

Figure 7: question-3 code

```
module tb_and_or_selector;

reg  [3:0] a;
reg  [3:0] b;
reg      sel;
wire [3:0] y;

and_or_selector uut (.a(a) , .b(b) , .sel(sel) , .y(y));

initial begin

    $display(" sel    a      b      y ");
    $monitor("  %b   %b   %b   %b", sel, a, b, y);

    //test inputs-1
    a = 4'b1010;
    b = 4'b0110;
    sel = 1'b0;#5;//for sel=0
    sel = 1'b1; #5;//for sel=1

   //test inputs-2
    a = 4'b0111;
    b = 4'b1110;
    sel = 1'b0; #5;//for sel=0
    sel = 1'b1; #5;//for sel=1

    $finish;

end
endmodule
```

Figure 8: testbench code

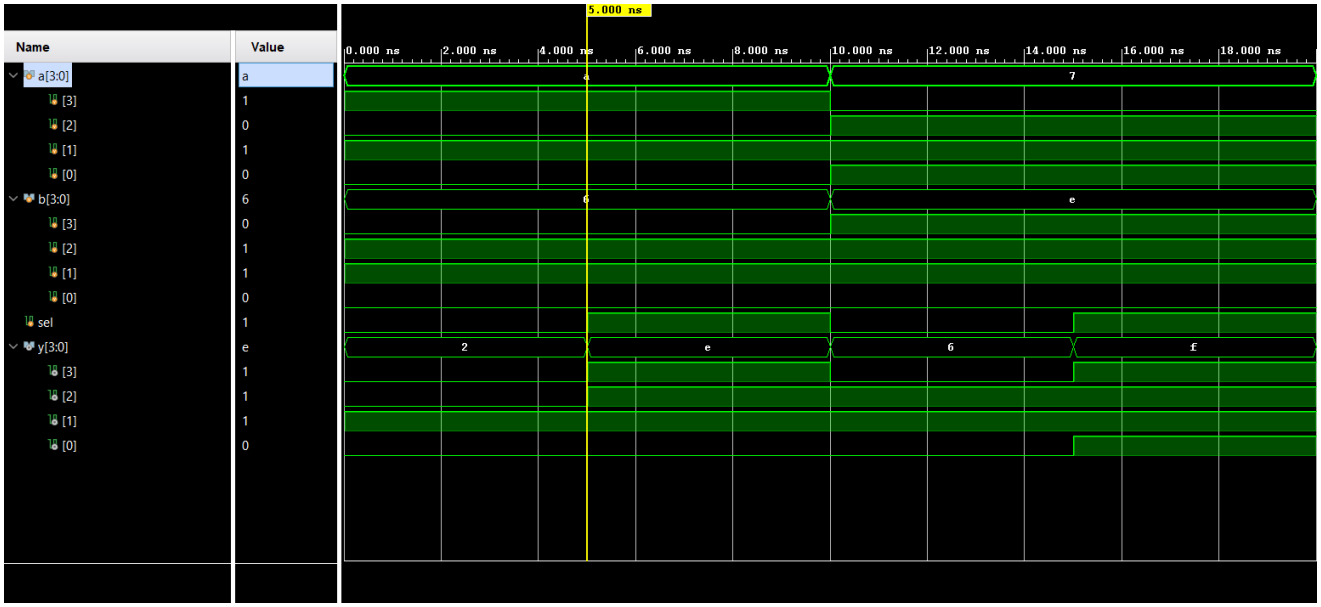

Figure 9: Waveform-3

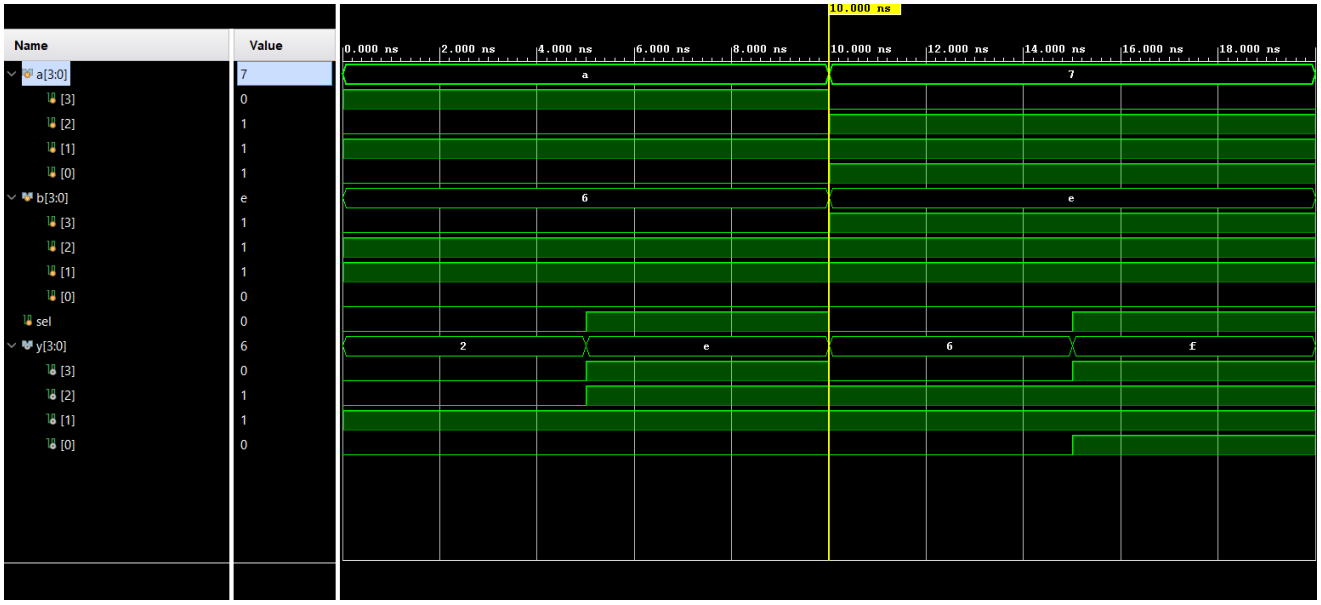

Figure 10: Waveform-3(1)

4 Question 4

Even Parity Generator

```
module even_parity(
    input [3:0] data,
    output parity
    );
    assign parity = data[3] ^ data[2] ^ data[1] ^ data[0];//XOR gives 1 when number of 1's is odd
endmodule
```

Figure 11: question-4 code

```
module tb_even_parity;

reg  [3:0] data;    // input
wire parity;        // output

even_parity uut ( .data(data) , .parity(parity));

initial begin

    $display(" data    parity ");
    $monitor(" %b      %b", data, parity);

  //testing with different cases
    data = 4'b0001; #5;
    data = 4'b0101; #5;
    data = 4'b1011; #5;
    data = 4'b1111; #5;

    $finish;

end

endmodule
```

Figure 12: testbench code

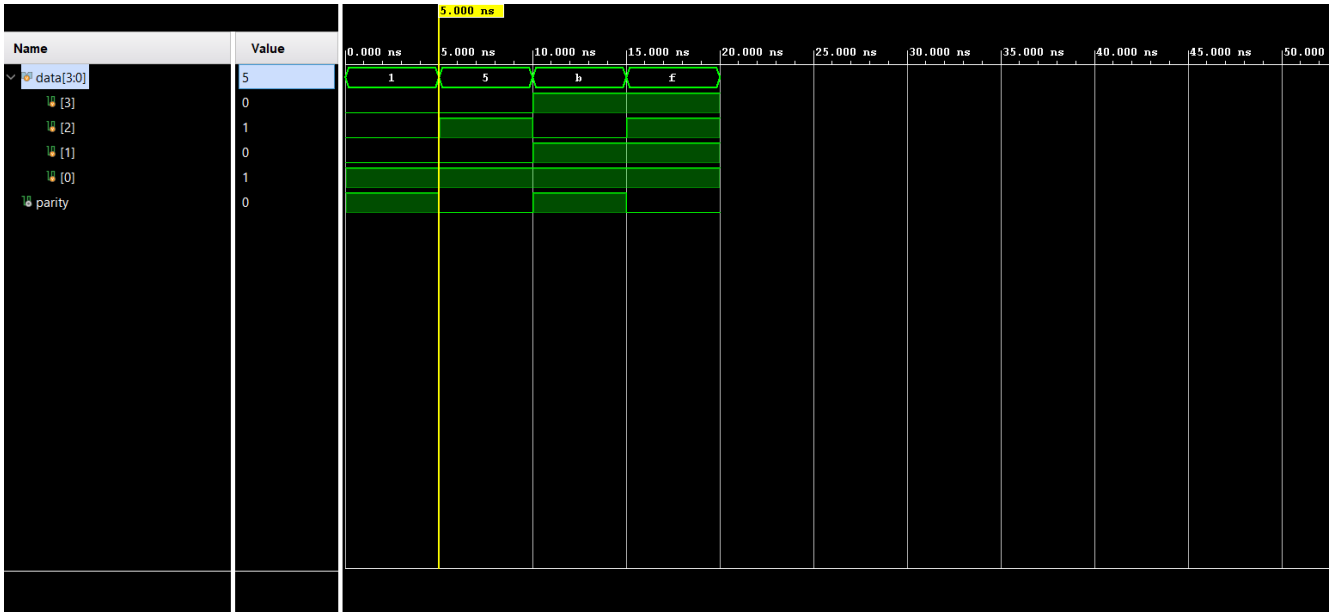

Figure 13: Waveform-4

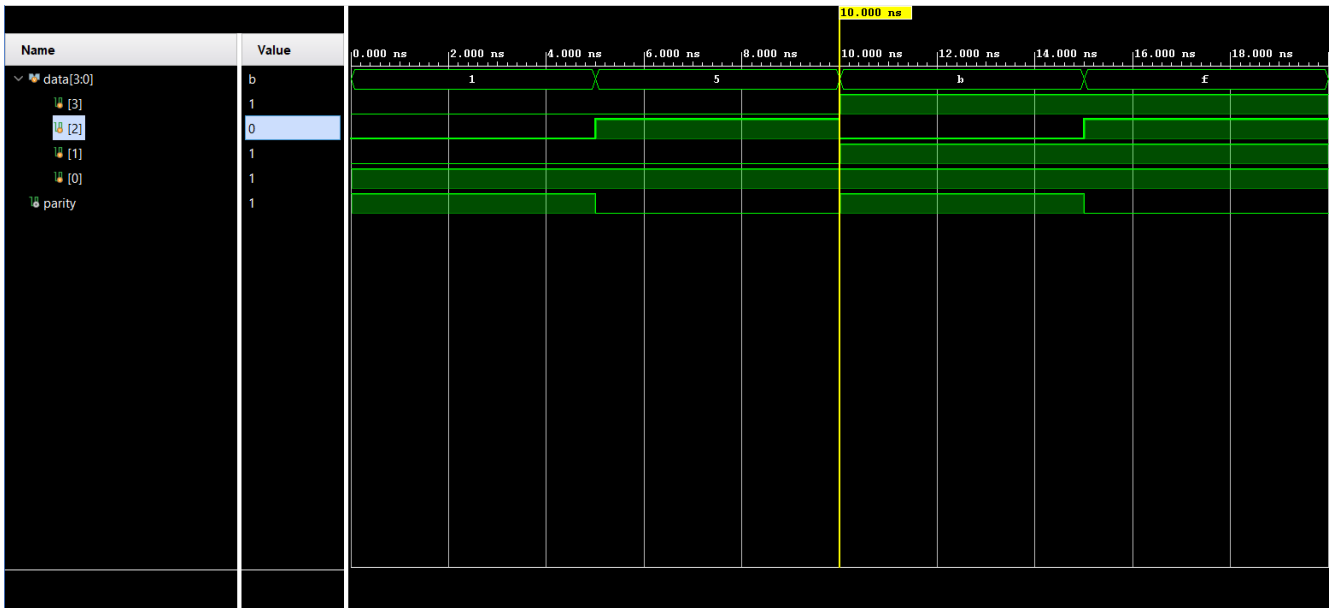

Figure 14: Waveform-4(1)

5 Question 5

Swapping Two bits

```
module swap_bits(
    input [3:0] in,
    output [3:0] out
    );
    assign out = {in[0] , in[2] , in[1] , in[3] } ; // swapping bit0 and bit 3
endmodule
```

Figure 15: question-5 code

```
module tb_swap_bits;

reg  [3:0] in;      // input
wire [3:0] out;     // output

swap_bits uut ( .in(in) , .out(out) );

initial begin

    $display("   in      out");
    $monitor("   %b      %b", in, out);

//testing for different cases
    in = 4'b0011; #5;
    in = 4'b0101; #5;
    in = 4'b0110; #5;
    in = 4'b1010; #5;

    $finish;

end
endmodule
```

Figure 16: testbench code

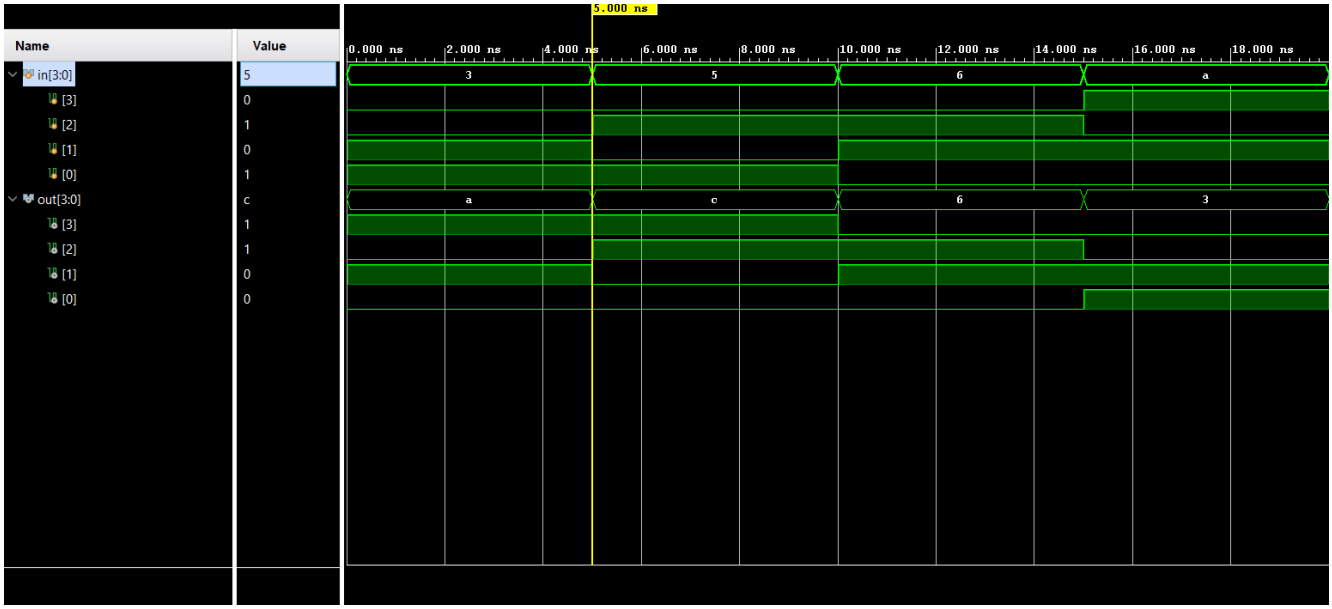

Figure 17: Waveform-5

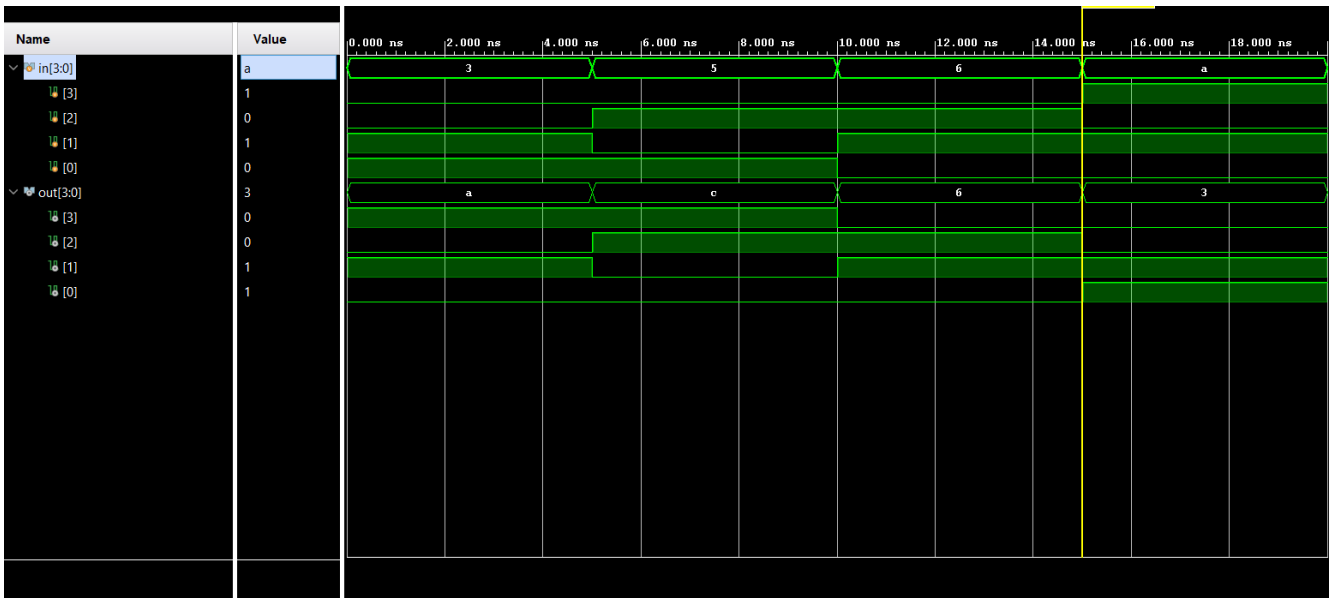

Figure 18: Waveform-5(1)